A valuation obstruction for 3×3 magic squares of squares, with a machine-checked proof

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Abstract

We study the open problem of whether a 3×3 magic square whose nine distinct entries are perfect squares exists. We reduce the problem to a purely additive condition on the set $D(e) = \{2xy : x^2 + y^2 = e^2\}$ attached to the center root e, and prove unconditionally that the center root of any such square must be divisible by at least two distinct primes $p \equiv 1 \pmod 4$: for centers $e = s\,p^a$ with a single such prime, the elements of $D(e)$ have pairwise distinct p-adic valuations, which is incompatible with the required additive structure. All results, including the reduction, are formally verified in Lean 4 with mathlib. We also report an exhaustive computation showing no magic square of squares has center entry below 10^{20}, and that Bremner's fully magic square with seven square entries is, up to scaling, the only one with center root below 10^9. Finally we describe a rigidity phenomenon that pins down the remaining two-prime case to explicit divisibility coincidences.

1 Introduction

Whether nine distinct perfect squares can form a fully magic 3×3 square is a well-known open problem, popularized by Martin Gardner's 1996 prize and studied extensively since; see Boyer's survey and problem collection [1]. The strongest results to date are computational (searches over magic sums and over arithmetic-progression parameters) together with structural constraints on hypothetical solutions due to Morgenstern, Roberts, Underwood and others [1].

Throughout, a *magic square of squares* is a 3×3 array of nine distinct perfect squares whose three rows, three columns and both diagonals share a common sum S. Summing the four lines through the center shows $S = 3E$ where $E = e^2$ is the center entry; we call e the *center root*.

Results

Theorem 1 (Reduction)**.** *Let $D(e) = \{2xy : x^2 + y^2 = e^2,\ 0 < x < y\}$. A magic square of squares with center e^2 exists if and only if there are u, v with u, v, $u + v$, $u - v \in D(e)$ (all nonzero, $u \neq v$); the square's entries are then e^2, $e^2 \pm u$, $e^2 \pm v$, $e^2 \pm (u + v)$, $e^2 \pm (u - v)$.*

Theorem 2 (Main theorem)**.** *If the center root e is divisible by at most one prime $p \equiv 1$* $\pmod 4$ *(to any power, and with arbitrary cofactor free of such primes), then no magic square* *of squares with center e^2 exists. Equivalently: the center root of any magic square of squares is divisible by at least two distinct primes $\equiv 1 \pmod 4$.*

Theorem 3 (Computations)**.** *There is no magic square of squares with center entry $e^2 < 10^{20}$. Moreover, every fully magic 3×3 square with exactly seven square entries and center root $e < 10^9$ is an integer scaling of Bremner's example with $e = 425$.*

All statements of Theorems 1 and 2, including Fermat's theorem that three squares in arithmetic progression have common difference divisible by 24, are formally verified in Lean 4 over mathlib; the library (thirteen files, no unproven assumptions) accompanies this paper.

2 The reduction

Every 3×3 magic square has the form $c \pm \{0, u, v, u+v, u-v\}$ arranged as

$$\begin{pmatrix} c+u & c-u-v & c+v \\ c-u+v & c & c+u-v \\ c-v & c+u+v & c-u \end{pmatrix},$$

in which all eight line sums equal $3c$ identically. Thus magic is free and the entire problem is the squareness of the nine cells.

Lemma 4. *For $c = e^2$, both $c - d$ and $c + d$ are squares if and only if $d \in D(e) \cup \{0\}$ (up to sign). Indeed if $x^2 + y^2 = e^2$ then $c - 2xy = (x-y)^2$ and $c + 2xy = (x+y)^2$; conversely if $c - d = A^2$ and $c + d = B^2$ then $A^2 + B^2 = 2e^2$, $A \equiv B \pmod 2$, and $x = \frac{B-A}{2}$, $y = \frac{B+A}{2}$ satisfy $x^2 + y^2 = e^2$, $2xy = d$.*

Theorem 1 follows by applying Lemma 4 to the four center lines. (Distinctness of the nine entries corresponds to $u, v, u \pm v \neq 0$ and $u \neq v$.)

3 Proof of the main theorem

Fix a prime $p \equiv 1 \pmod 4$, write $p = \pi\bar{\pi}$ in $\mathbb{Z}[i]$ with $\pi = A + Bi$, $A^2 + B^2 = p$, and let $e = s\,p^a$ with $a \geq 0$ and s free of primes $\equiv 1 \pmod 4$.

Lemma 5 (Rigid part)**.** *Every $z \in \mathbb{Z}[i]$ with $N(z) = s^2$ is a unit multiple of s.*

Proof. Induct on the least prime factor q of s. If $q \equiv 3 \pmod 4$ then q is inert; $q \mid N(z) = z\bar{z}$ gives $q \mid z$, and we divide. If $q = 2$, then $1 + i$ divides z twice (as $2 \mid N(z)$ twice), and $(1+i)^2 = 2i$ is 2 times a unit. ☐

Lemma 6 (Classification)**.** *Every $z \in \mathbb{Z}[i]$ with $N(z) = s^2 p^n$ equals $u\,s\,\pi^j \bar{\pi}^{n-j}$ for a unit u and some $0 \leq j \leq n$.*

Proof. Induct on n: for $n \geq 1$, $\pi \mid p \mid z\bar{z}$, so $\pi \mid z$ or $\bar{\pi} \mid z$; divide and recurse, finishing with Lemma 5. ☐

Lemma 7 (Bridge)**.** *For every $m \geq 1$, $p \nmid \mathrm{Im}(\pi^m)$.*

Proof. Pick $\omega \in \mathbb{Z}/p$ with $\omega^2 = -1$ (possible as $p \equiv 1 \bmod 4$) and consider the ring maps $\varphi_\pm : \mathbb{Z}[i] \to \mathbb{Z}/p,\ i \mapsto \pm\omega$. If $p \mid \operatorname{Im}(\pi^m)$ then, since $\operatorname{Re}(\pi^m)^2 + \operatorname{Im}(\pi^m)^2 = p^m$, also $p \mid \operatorname{Re}(\pi^m)$, so $\varphi_\pm(\pi)^m = 0$; as $\mathbb{Z}/p$ is a field, $\varphi_\pm(\pi) = 0$, i.e. $A \pm B\omega \equiv 0$. Adding and subtracting gives $p \mid A$, $p \mid B$, contradicting $A^2 + B^2 = p$. $\square$

Lemma 8 (Representation structure)**.** *Let $x^2 + y^2 = e^2 = s^2p^{2a}$ with $xy \neq 0$. Then there are $1 \le t \le a$ and $\varepsilon \in \{\pm 1\}$ with*

$$2xy \;=\; \varepsilon\, p^{\,2(a-t)}\left(s^2 \operatorname{Im}\pi^{4t}\right), \qquad p \nmid s^2 \operatorname{Im}\pi^{4t}.$$

In particular $\nu_p(2xy) = 2(a-t)$, and distinct elements of $D(e)$ have pairwise distinct p-adic valuations $\{0, 2, \ldots, 2a-2\}$.

Proof. Write $z = x + yi$; by Lemma 6, $z = us\pi^j\bar{\pi}^{2a-j}$. Then $2xy = \operatorname{Im}(z^2)$ and $z^2 = u^2s^2\pi^{2j}\bar{\pi}^{2(2a-j)}$ with $u^2 = \pm 1$. If $j = a$ the product is real and $xy = 0$. Otherwise, with $t = |j-a|$, $\pi^{2j}\bar{\pi}^{2(2a-j)} = p^{2(a-t)}\cdot\pi^{\pm 4t}$, and $\operatorname{Im}(\bar{\pi}^{4t}) = -\operatorname{Im}(\pi^{4t})$; absorb signs into ε. Lemma 7 and $p \nmid s$ give the non-divisibility. $\square$

Proof of Theorem 2. Suppose $u, v, u+v, u-v \in D(e)$ as in Theorem 1, and give each the form of Lemma 8 with parameters (t_i, ε_i) and constants $c_t = s^2 \operatorname{Im}\pi^{4t}$. If $t_1 = t_2$ then $u = \pm v$, excluded. Otherwise $\nu_p(u) \neq \nu_p(v)$; writing $u \pm v = p^{\,\min}\cdot(\text{bracket})$ where the bracket is $\varepsilon_i c_{t_i} + (\text{multiple of } p)$, the bracket is prime to p. Comparing with the exact forms of $u+v$ and $u-v$ and using uniqueness of exact p-power extraction, $2(a-t_3) = 2(a-t_4) = \min$, so $t_3 = t_4$ and $u + v = \pm(u - v)$, forcing $u = 0$ or $v = 0$. Contradiction. $\square$

Remark 9. The counting argument (four center lines require four representations) already forces two distinct primes *or* $p^4 \mid e$; the valuation argument closes the prime-power loophole entirely and is what we formalize.

4 Computations

Enumerating candidate centers directly through Theorem 1 is far cheaper than enumerating magic sums: factoring e in windows and generating $D(e)$ from the Gaussian factorization checks all $e < 10^{10}$ (center entries to 10^{20}) in minutes on a desktop; no configuration $u, v, u \pm v \in D(e)$ exists in that range. The same machinery classifies near-misses: configurations with two of the four differences in $D(e)$ and the other two "half in" (one square cell each) are exactly the fully magic squares with seven square entries, of which Bremner's $e = 425$ example was the only one known. Scanning all $e < 10^9$ finds 2,352,941 such configurations — every single one an integer scaling of $e = 425$, and none with eight square entries.

5 The minimal two-prime case

Theorem 10. *There is no magic square of squares whose center root is $s\,pq$, where $p \neq q$ are primes $\equiv 1 \pmod 4$, each to the first power, and s is free of such primes.*

Proof sketch. Here $D(e)$ has exactly four elements: the interior conjugate pair $m, m' = s^2|\operatorname{Im}(\pi^4\chi^{\pm 4})|$ and the axis elements $X = s^2q^2|\operatorname{Im}\pi^4|$, $Y = s^2p^2|\operatorname{Im}\chi^4|$. Since $u, v, u+v, u-v$

are four distinct elements of $D(e)$, they exhaust it. Write $\pi^4 = R_A + I_A i$, $\chi^4 = R_B + I_B i$; recall R_A, R_B are odd, $4 \mid I_A, I_B$, and by Lemma 7 $p \nmid I_A$, $q \nmid I_B$ (and $I_A, I_B \neq 0$). Enumerate the placements of $\{X, Y\}$ among $\{u, v, u+v, u-v\}$:

(1) $\{u, v\} = \{X, Y\}$. Then $\{u \pm v\} = \{m, m'\}$ and the conjugate-pair identities force $\{u, v\} = \{s^2|R_B I_A|,\ s^2|I_B R_A|\}$. Matching against X, Y and multiplying the two resulting equations gives $|R_A R_B| = p^2 q^2$; but $|R_A| < p^2$ and $|R_B| < q^2$ strictly (as $I_A, I_B \neq 0$), a contradiction. The crossed matching forces $|R_A| = p^2$ outright, equally impossible.

(2) $\{u, v\} = \{m, m'\}$. Then $\{u \pm v\} = \{X, Y\}$ and $u+v,\ u-v \in \{2s^2|R_B I_A|,\ 2s^2|I_B R_A|\}$; matching yields either $2|R_B| = q^2$ or $4|R_A R_B| = p^2 q^2$, both impossible by parity.

(3) mixed placements. One axis element lies in $\{u, v\}$, the other in $\{u \pm v\}$. Each of the 64 sign/placement sub-cases yields two linear equations in $W = R_A I_B$, $Z = I_A R_B$; solving and substituting into the norm relations $(W/I_B)^2 + I_A^2 = p^4$, $(Z/I_A)^2 + I_B^2 = q^4$ leaves a difference that factors, in every sub-case, into terms from the list

$$(I_A q^2 \pm I_B p^2), \quad (2I_A q^2 \pm I_B p^2), \quad (I_A q^2 \pm 2I_B p^2), \quad 3I_A q^2, \quad 3I_B p^2.$$

The all-positive combinations cannot vanish; the difference combinations vanish only if $I_A q^2 \in \{I_B p^2,\ 2I_B p^2\}$ or $I_B p^2 = 2I_A q^2$, each of which equates a quantity prime to p with one divisible by p^2 (Lemma 7). Hence no sub-case admits a solution. $\square$

Remark 11. Bremner's seven-square-entry example has center root $425 = 5^2 \cdot 17$, just outside the hypothesis (exponent 2): consistent with Theorem 10, the near-miss lives at the smallest center type the theorem does not reach.

6 Center roots $s\,p^2 q$: Bremner's type

Theorem 12. *There is no magic square of squares whose center root is $s\,p^2 q$, where $p \neq q$ are primes $\equiv 1 \pmod 4$ and s is free of such primes.*

Here $|D(e)| = 7$: writing $\pi^4 = R_A + I_A i$, $\chi^4 = R_B + I_B i$ and $P = p^2$, $Q = q^2$, the elements are (up to the global factor s^2)

$$P\,|\mathrm{Im}(\pi^4 \chi^{\pm 4})|, \quad |\mathrm{Im}(\pi^8 \chi^{\pm 4})|, \quad PQ\,|I_A|, \quad Q\,|\mathrm{Im}\,\pi^8|, \quad P^2|I_B|.$$

All seven are *linear* in (R_B, I_B) over $\mathbb{Q}(R_A, I_A, P, Q)$. The four differences $u, v, u+v, u-v$ occupy four distinct elements; for each of the 840 placements and each resolution of the absolute values (6,720 linear systems in all), solving the two placement equations for (R_B, I_B) and substituting into $R_B^2 + I_B^2 = Q^2$ leaves one polynomial relation in (R_A, I_A, P, Q), which must vanish. A computer-assisted but fully rigorous sweep (scripts accompany this paper) shows every relation is impossible:

- 339 of the 343 distinct relation classes die by combinations of: parity (R_A, R_B, P, Q odd, $8 \mid I_A, I_B$); the p-adic clash (ν_p of the two sides differ, using $p \nmid R_A I_A$ and Lemma 7); congruence obstructions modulo 16, 32 and small odd moduli on the curve $R_A^2 + I_A^2 = P^2$; the strict size pinch $|R_A|, |I_A| < P$; and elimination of I_A via the norm followed by factorization. The finitely many exceptional primes ($\ell \in \{5, 13, 17\}$, arising from small constants in the eliminants) are closed by exact evaluation at the actual π^4.

- 4 classes reduce to homogeneous quintics/sextics in (R_A, P); a rational root R_A/P in lowest terms would need $P = p^2$ to divide a leading coefficient in $\{\pm 8, \pm 16, \pm 32, \pm 40\}$ — impossible.
- 4 classes are tautological (the norm residual vanishes identically); there the solved values are $(R_B, I_B) = \pm(Q/P)(R_A, -I_A)$, and integrality forces $p^2 \mid q^2$.

The 256 distinct determinants of the linear systems are proven nonzero by the same mechanisms, so no singular case escapes. □

Remark 13. Bremner's fully magic square with seven square entries has center root $425 = 5^2 \cdot 17$, of exactly this type: Theorem 12 shows the missing two entries can never be repaired at *any* center of that shape. Since the seven D-elements are linear in (R_B, I_B) for every $b = 1$ exponent pattern $s\,p^a q$, the same pipeline applies to all $(a, 1)$ cases; the case count grows with a, and a uniform-in-a argument is the natural next objective.

Theorem 14. *There is no magic square of squares whose center root is $s\,p^3 q$ ($p \neq q$ primes $\equiv 1 \bmod 4$, s free of such primes).*

For $a = 3$ the difference set has ten elements, still linear in (R_B, I_B); the sweep comprises 64,512 leaf systems in 3,767 relation classes. All 1,271 nonsingular classes die by the Theorem 12 mechanisms; the 2,496 singular classes (rank-deficient placements, which first appear at $a = 3$ through the three constant axis elements) die through their consistency relations by the same kills; and all 1,080 determinant classes are proven nonzero, so no singular locus escapes. The certificates for $(a, b) = (1, 1), (2, 1), (3, 1)$ share one pipeline, and every two-prime exponent pattern (a, b) is decidable in this sense — for $b \geq 2$ the leaf systems become zero-dimensional quadratic rather than linear, since $\chi^8, \chi^{12}, \ldots$ are polynomials in (R_B, I_B). The open problems are uniformity in the exponents and the ≥ 3-prime landscape, where the leaf systems acquire positive dimension.

7 The two-prime frontier

Toward uniformity in the exponent

The certificates for $(a, b) = (1, 1), (2, 1), (3, 1)$ suggest a uniform theorem for all $s\,p^a q$. Abstracting the levels of $D(e)$ into free circle variables linked by a chain map (the enriched relaxation), 95.8% of the 9,216 uniform two-level leaf systems die by mechanisms independent of a — positivity pinches, parities, p-adic patterns and congruence obstructions. The residue is exactly 24 relations, independent of a, involving two π-powers π^{4j}, $\pi^{4(j+\Delta)}$. A complete weight-regime analysis shows that their phase-aligned leading forms cancel identically: the obstruction survives every first-order p-adic tool. Resolving the residue requires second-order lifts along the π-power orbit, or remains a per-(j, Δ) finite check; it is the precise uniform incarnation of the divisibility-coincidence ladder behind Bremner's near-miss.

For $e = s\,p^a q^b$ with two distinct primes $\equiv 1 \pmod 4$, the valuation pairs (ν_p, ν_q) organize $D(e)$ into *conjugate pairs* $\{d_{j,k}, d_{j,-k}\}$ of multiplicity exactly two off the axes, and axis singletons. If $\{u+v, u-v\}$ is a conjugate pair $\{d, d'\}$ arising from $A = \pi^{4j}$, $B = \chi^{4k}$, then

$$d + d' = 2\,\mathrm{scale} \cdot |\,\mathrm{Re}\,B\ \mathrm{Im}\,A|, \qquad d - d' = 2\,\mathrm{scale} \cdot |\,\mathrm{Im}\,B\ \mathrm{Re}\,A|,$$

so u and v are *completely determined* — and their valuation pair coincides with that of $\{d, d'\}$, forcing $u, v \in \{d, d'\}$, which is absurd. Consequently every hypothetical two-prime solution requires an explicit divisibility coincidence ($q^t \mid \operatorname{Re} \pi^{4j}$, $p^t \mid \operatorname{Re} \chi^{4k}$, or a mod-$p$ or mod-q cancellation in $u \pm v$). These coincidences recur along arithmetic progressions of exponents, so closing them requires exact-magnitude input beyond valuations; we leave this as the principal open direction, noting that together with Theorem 2 it would force every solution's center root into three or more primes $\equiv 1 \pmod 4$.

References

[1] C. Boyer, *Latest research on the "3x3 magic square of squares" problem*, `http://www.multimagie.com/English/SquaresOfSquaresSearch.htm`.

[2] A. Bremner, *On squares of squares II*, Acta Arith. 99 (2001), 289–308.

[3] M. Gardner, *The magic of 3x3*, Quantum 6 (1996), 24–26.